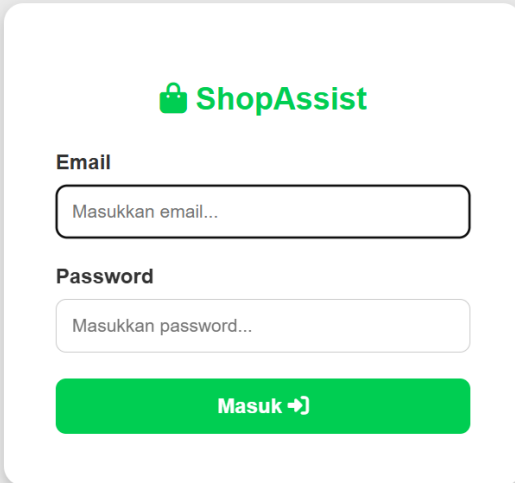


Documentation

Email and Password

Email and password are used to identify users for security purposes. Use the registered email and password.

A login form for ShopAssist. At the top is the ShopAssist logo, which consists of a green shopping bag icon followed by the text 'ShopAssist' in green. Below the logo, there are two input fields. The first is labeled 'Email' and contains the placeholder text 'Masukkan email...'. The second is labeled 'Password' and contains the placeholder text 'Masukkan password...'. Below these fields is a green button with the text 'Masuk ➔' in white.

Product Data Acquisition and Preprocessing

Product Data Acquisition is the systematic collection of information (attributes, specifications, prices, inventory) about products from various sources, such as supplier feeds and APIs. Preprocessing transforms the often-chaotic raw data into a clean, structured dataset, ready for use by e-commerce platforms, machine learning, or analytics.

1. Data Acquisition Methods

Web Scraping and Crawling is used to extract product details directly price and availability from websites.

2. Data Preprocessing Stages

Data Cleaning is used to handle missing attributes for example, inferring missing weights, removing duplicates, and standardizing categorical values.

3. Key Objectives

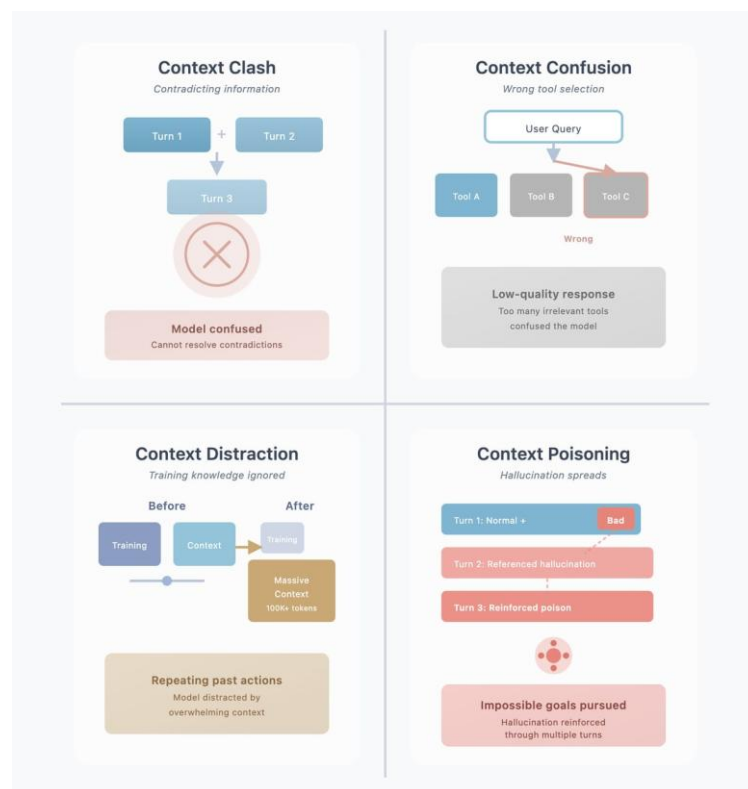
The goal will be to prepare data sets for search engine indexing or AI recommendations.

Structured Prompt Design for Product Comparison

Structured Prompts are used to construct product comparison questions for Large Language Models (LLMs), which require a shift from vague, conversational questions to a specific, targeted framework. Structured prompt design uses a specific architectural framework to guide LLMs in generating consistent, accurate, and actionable product comparison content. Structured prompts for product comparisons use predefined constraints to force the AI to perform objective and in-depth analysis, detailing roles, context, and formatting, while avoiding generic summaries to produce accurate, high-quality comparison matrices and content.

Context Pruning for Long Product Descriptions

Context Pruning for Long Product Descriptions is an artificial intelligence (AI) technique for trimming repetitive, irrelevant, or low-priority information from product documents before sending them to a Large Language Model (LLM). This dramatically improves e-commerce accuracy, cuts token costs, and reduces application response latency.



A token is the smallest unit that can be processed in an LLM model. The model can understand the concept that it inputs tokens, not words. This is because context is used to construct sentences. Tokenization is the process of replacing sensitive data with a digital representation or unique code (token). This technology is used to safeguard data processing in artificial intelligence (AI) and digitize asset ownership.

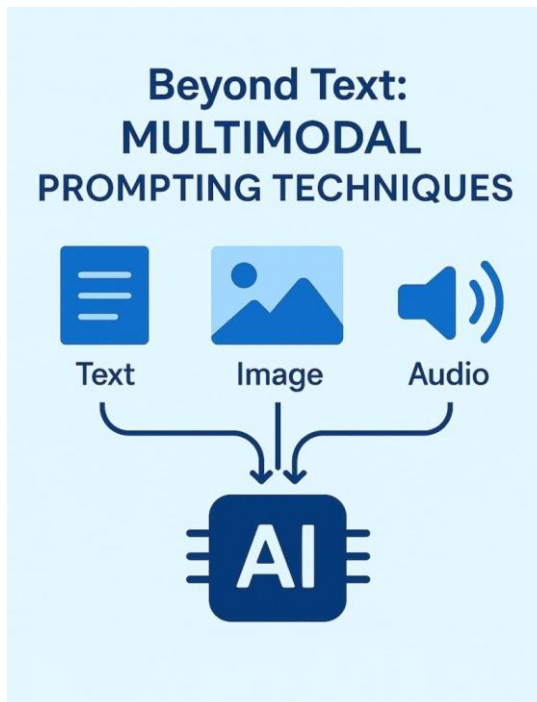
- **Text processing:** In AI like ChatGPT or Gemini, tokenization is the process of breaking down long sentences or texts into smaller pieces of words or syllables called tokens, so that they can be read and processed by machines.
- **Data analysis:** Helps computer systems understand human language patterns, translate them, and generate responses efficiently.

If an LLM model were to understand a single sentence when fed a full sentence, it would require a significant amount of memory. Because the system would use more memory, and ultimately, the more memory it uses, the greater its resources and the longer its computation time.

Context in an AI prompt is all the information the model inputs into an agent before generating a response. Context helps the model understand the prompt, and the model will respond by providing an answer based on the context inputted into the model.

Multimodal Prompting for Image-Based Product Understanding

Multimodality is important because it is a system that combines text and images. Images can help AI models understand visual data, or visual information can be used as a reference, making it easier for AI models to interpret complex scenarios that require additional information such as tables, graphs, image-based prompts, and visual prompts. This will improve the response. If multimodality functions rationally and analytically, multimodality will emerge if the model processes image-based content and visuals by integrating text-based instructions written in the prompt. This allows the text to be combined with additional image-based information to produce a response. If an AI model reads an image, the input and visual input must provide clear instructions so the AI model can understand the image and the information contained within it. The best visual-based prompts must include the context of the image being represented, for example, by inputting an image and writing it in the prompt. The output can be structured in the form of a point-based table or a list. For example, what objects are detected in the image, or in table form.



Conversational Shopping Assistant with Multi-Turn Context

A multi-turn, context-driven conversational shopping assistant is an AI agent designed to remember previous questions and user preferences throughout a conversation. Instead of treating each request as a new interaction, it builds an ongoing dialogue to refine product recommendations, handle questions, and seamlessly complete complex shopping tasks.

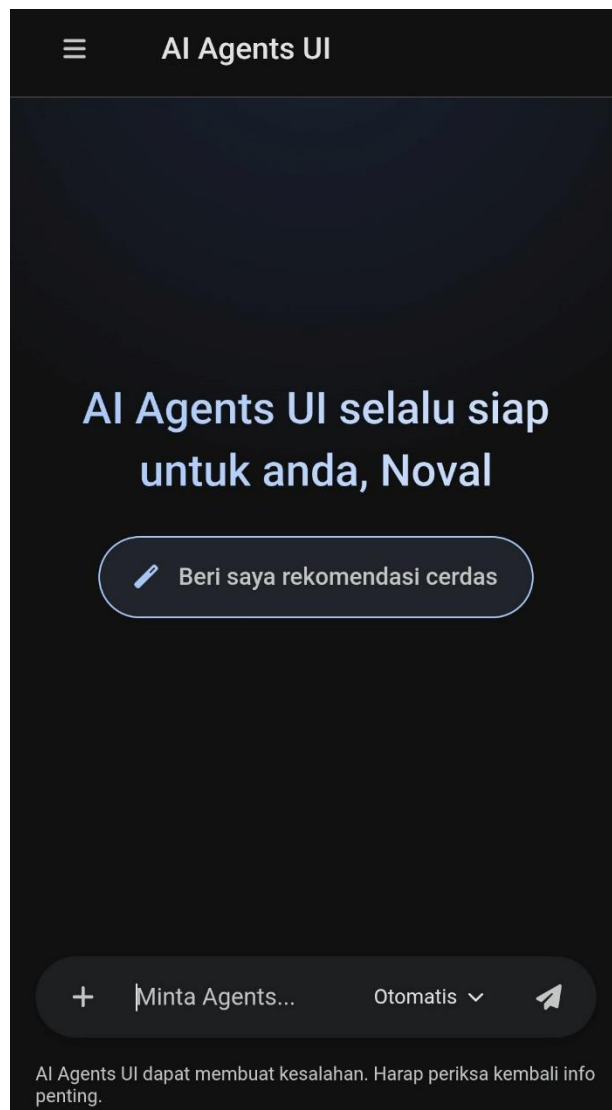
Main Abilities:

- **Context Retention:** AI remembers size, color, budget, or previous items discussed in chat.
- **Dynamic Refinement:** Users can gradually narrow down choices without having to restate basic parameters.
- **Conversational Commerce:** Integrates with catalog and inventory in the backend system to provide real-time pricing and availability.
- **Multi-Step Troubleshooting Guide:** Guides users through troubleshooting product issues, sizing, or returns without the need to be redirected to a human representative.



Mobile

Mobile refers to mobile devices such as smartphones or tablets. Mobile web, or mobile web, refers to website pages specifically designed and optimized for access via mobile devices such as smartphones and tablets.



Mobile web applications are websites optimized for mobile devices and specifically designed to function like real mobile apps. Web-based applications are optimized to look and feel like mobile apps when opened through a mobile browser. These applications do not require installation on internal memory.